Ngee Ann Polytechnic
School of Engineering (SoE)

Project 1: Routing with OSPF and configure switching for VLANs with VLSM

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Phase 1: Company & Network Requirements

Summary of Company and Network Requirements

As a small company that specialises in developing high-speed wireless products, the XYZ Research Company requires a corporate network tailored to fit their networking needs. As such, they approached us, the SSN Network Consulting Group, to assist them in designing a network which would meet their needs. The purpose of this document is to cover the findings and results of our project.

To begin, we will start by sharing about the physical layout of the company. The main office occupies two buildings in Sydney, with one building solely for the Administration Group. The other building is for the Sales and Marketing Group, and the larger Research and Development Group. The main building comprises three floors; The Research Group and the Sales and Marketing Group will each have employees located on all three floors. The XYZ Research Company also has a Sales Branch Office located in Melbourne.

The company plans to implement a wired network that is capable of supporting 100% growth over the next five to ten years. This is our task; To design, implement and fully document the XYZ Research Company network.

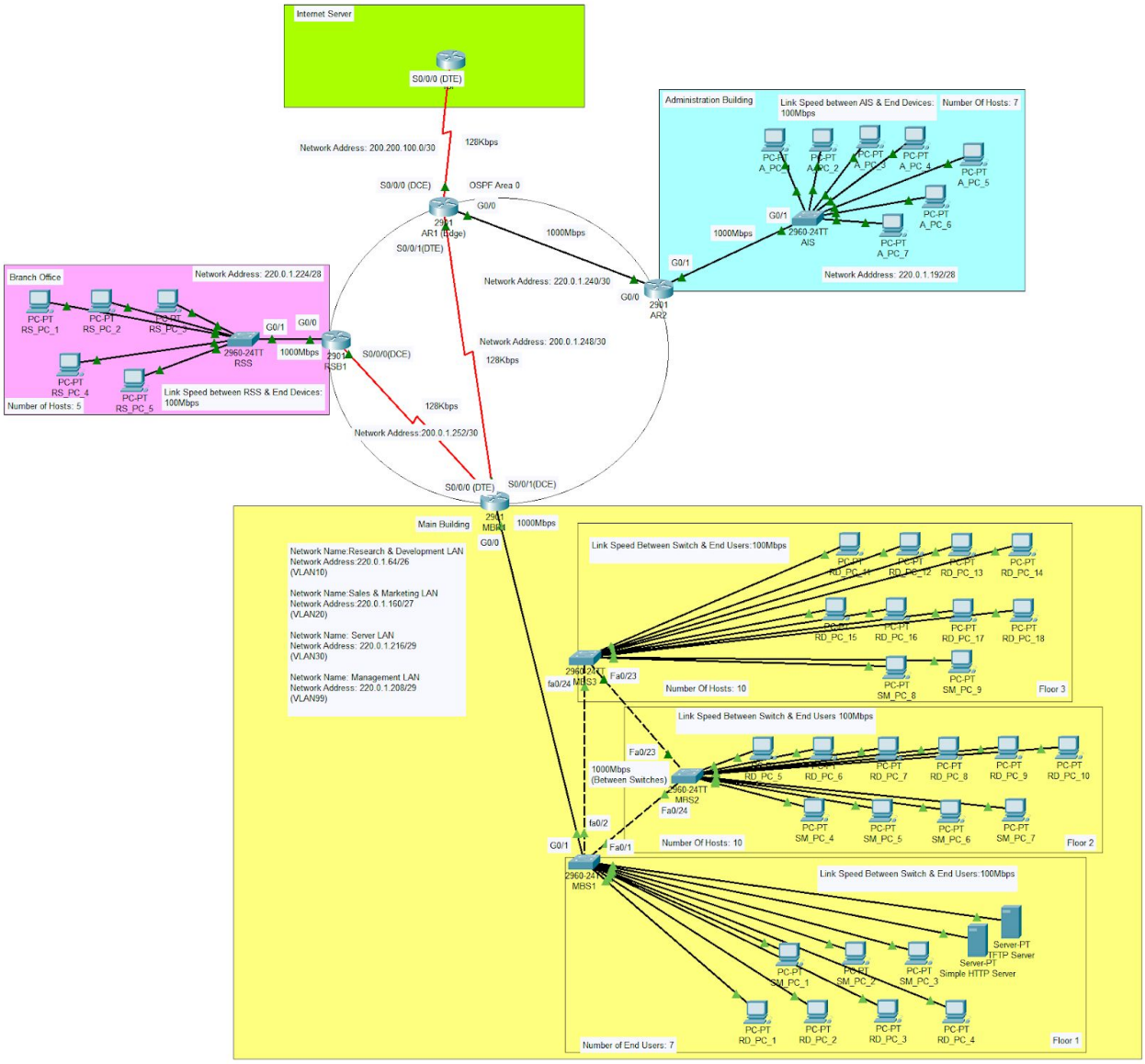
In addition to this formal report, we have also designed and tested a prototype of the network to verify that it will meet the company's needs. Various other documents have also been prepared, containing detailed information of the design and layout of the network. This ensures that a suitable third party can install and configure the network without any additional hassle.

The XYZ Research Company has provided a partially completed logical diagram of the required network. In addition, they have also provided us with a list of the other requirements to be factored into the design of this network.

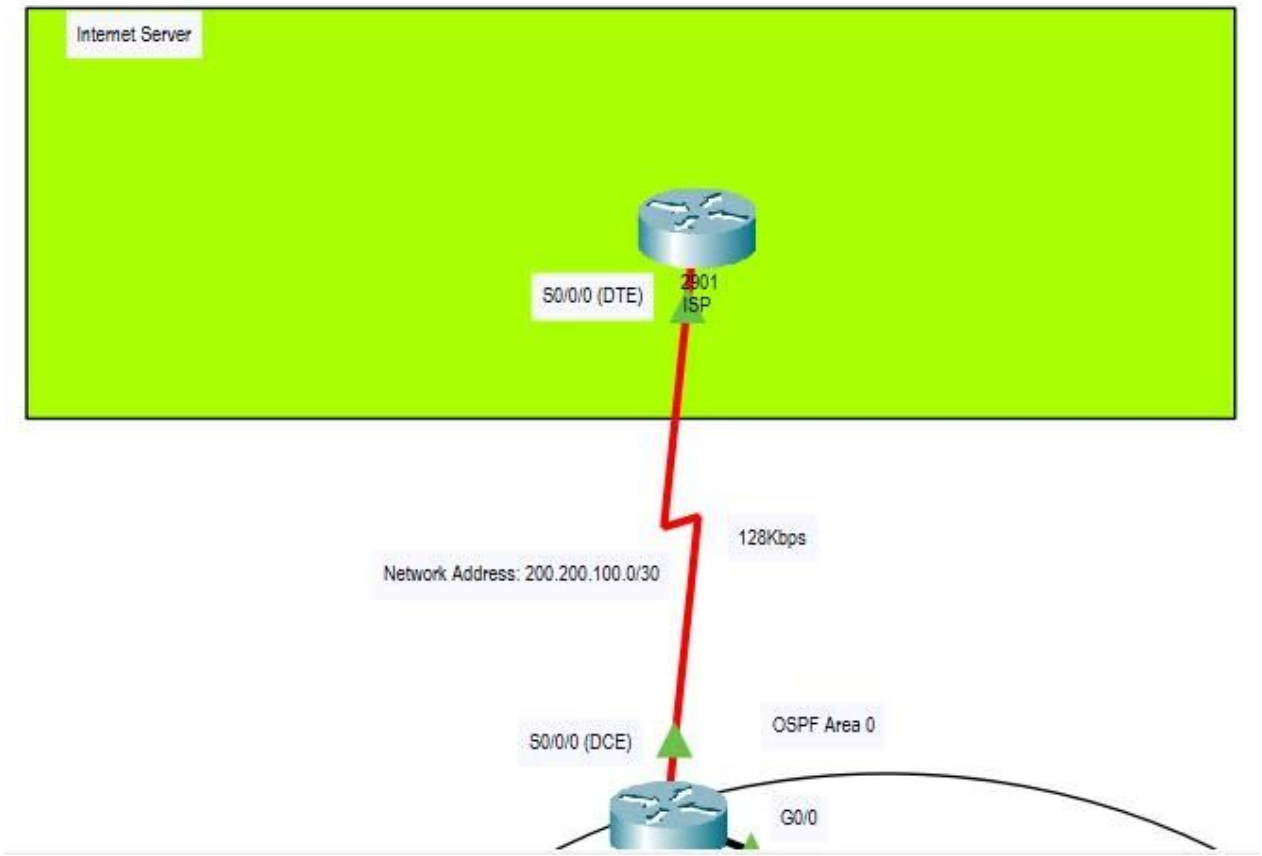
The requirements are as follows:

- 18 employees in the Research and Development group
- 9 employees in the Sales and Marketing group
- 7 employees in the Administration group
- 5 employees in the remote sales office
- Lifetime Max of 5 servers on separate subnet, regardless of company growth
- Use subnet 200.200.100.0/30 for connection to the Internet router
- Use public class C network 223.0.0.0 for internal addressing
- Use VLSM for IP addressing
- Expect 100% growth of current IP requirements when determining size of subnets
- All networking devices must have IP addresses

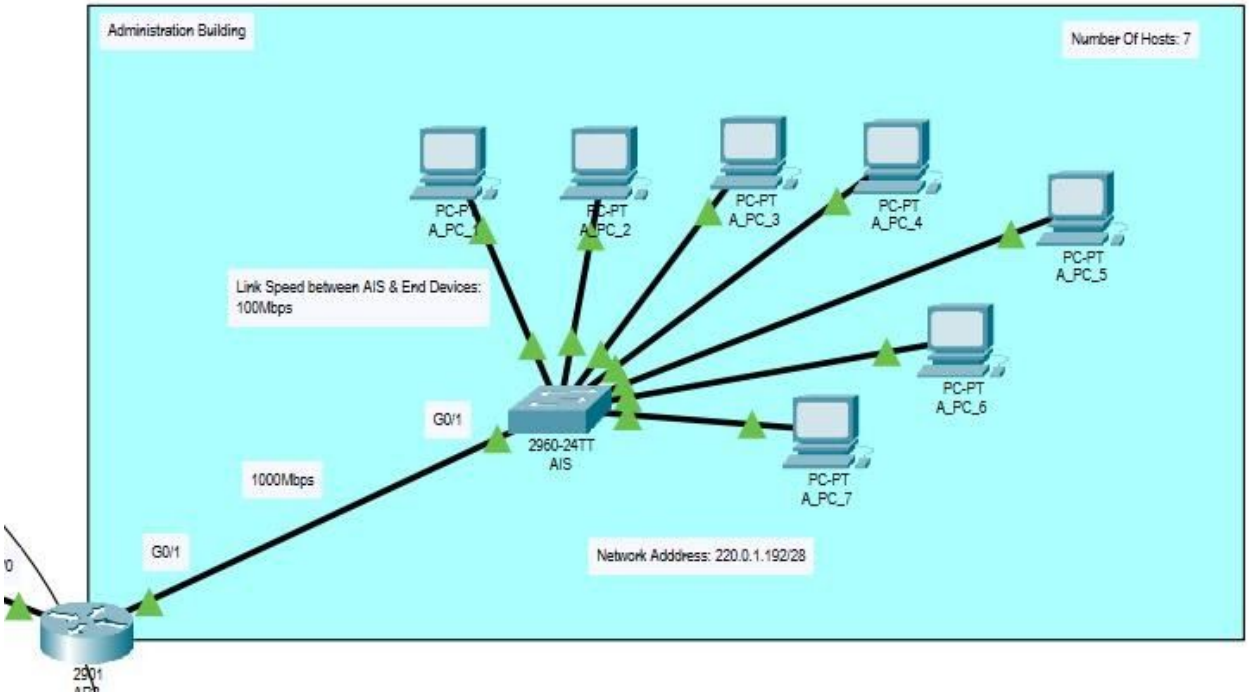
Logical Topology Diagram (Full Topology)



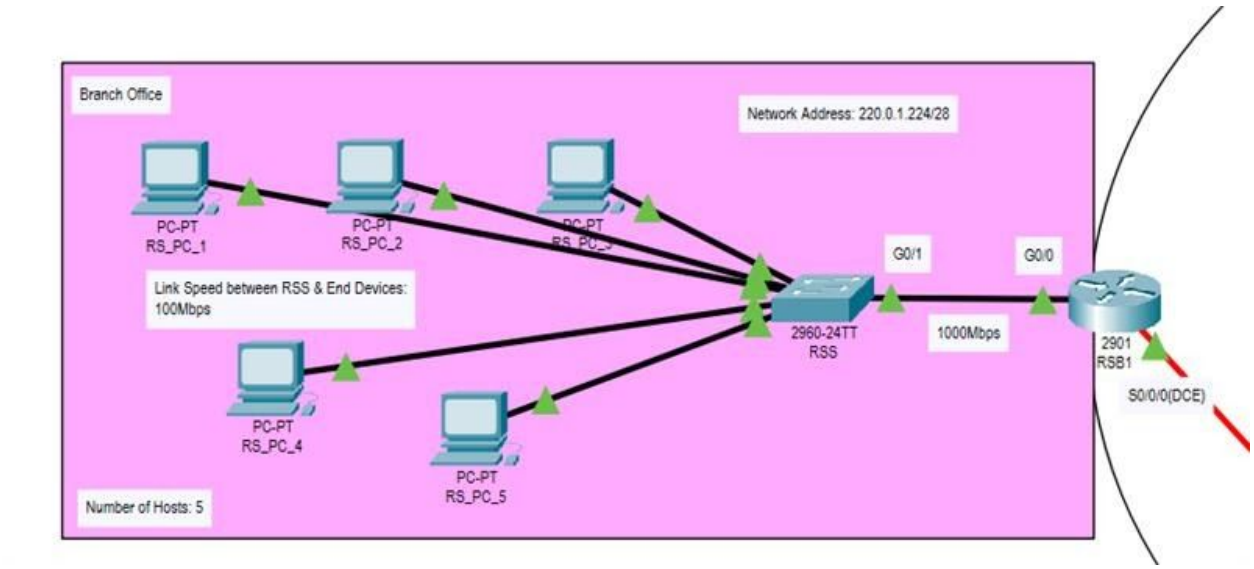
Logical Topology Diagram (ISP)



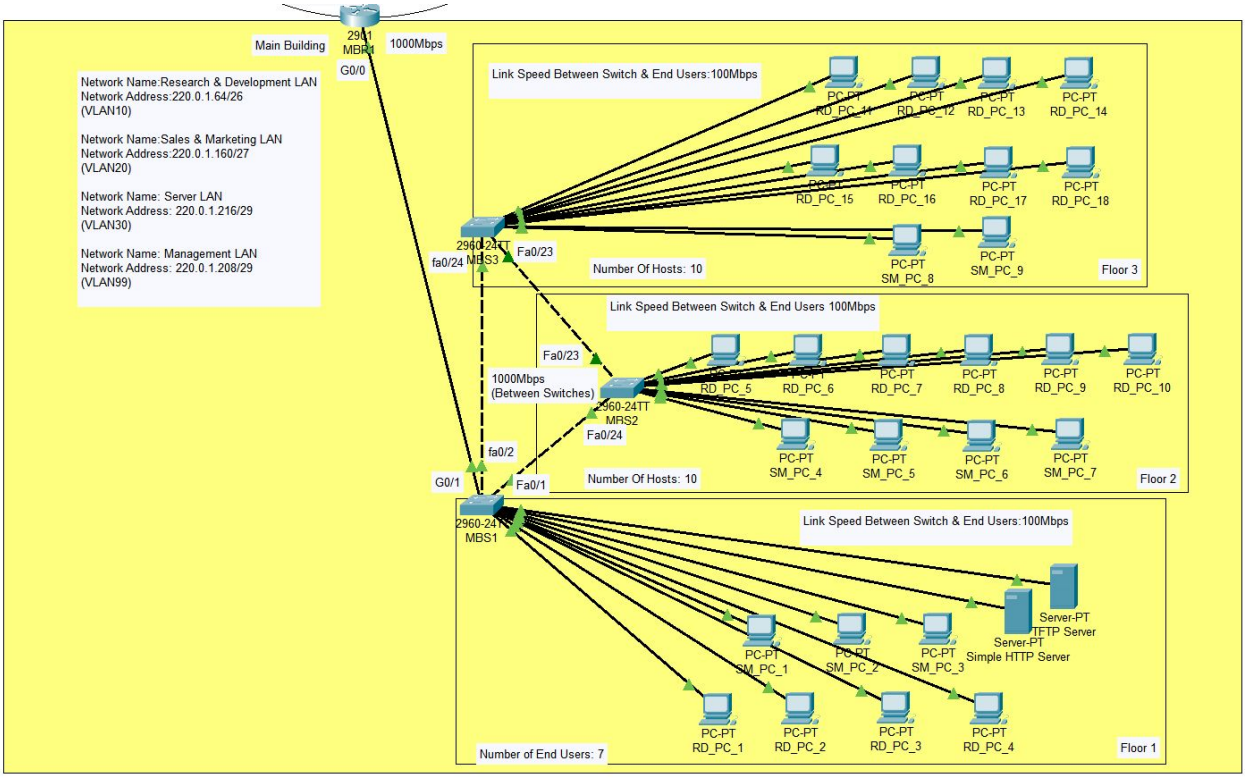
Logical Topology Diagram (Administration Building)



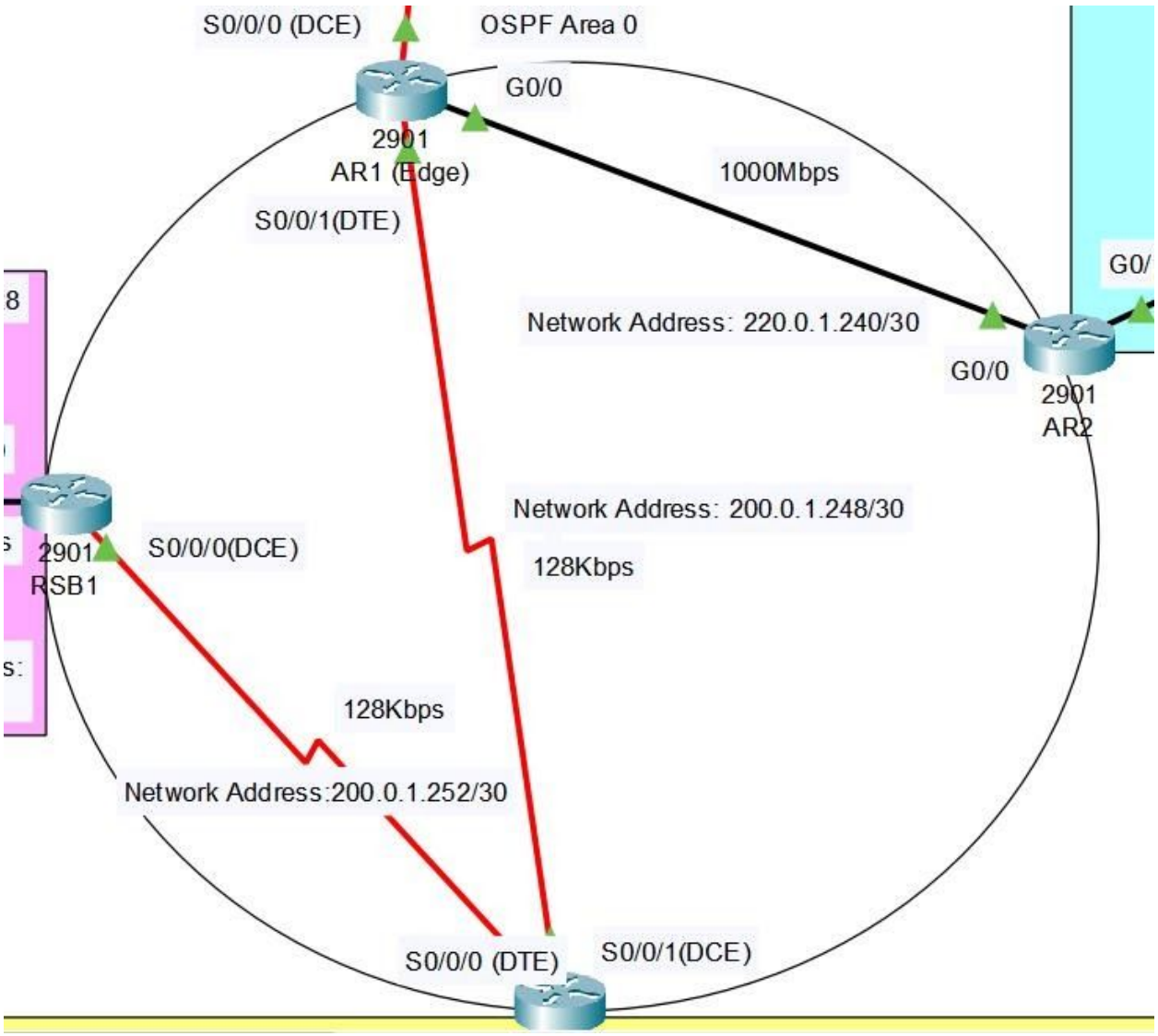
Logical Topology Diagram (Branch Office)



Logical Topology Diagram (Main Building)



Logical Topology Diagram (WAN)



IP Address & VLSM Implementation

As part of the Company's expectations, the use of VLSM design will be implemented in order to maximise the use of IP addresses. Due to this, we have calculated and designed various subnets that meet the requirements of the network. All of this information has been translated into a series of tables as shown below. We have also identified the excess subnets produced by our VLSM design, which will be included in the tables as well.

We will use public class C network 220.0.1.0/24 along with VLSM for internal addressing. This network will be subnetted based on the number of hosts needed per subnet accounting for 100% growth, while maximizing the number of remaining subnets available for future use. The public 200.200.100.0/30 network will be used for the connection between XYZ's edge router and the ISP. Other IP addresses such as 1.1.1.1, 2.2.2.2, 2.1.1.1, and 3.1.1.1 will be used on loopback interfaces on the router's used for its OSPF process (Explained in phase 3).

IP Address Range Table (220.0.1.0/24)

Number of host addresses required	Network Address (Host Range)	Subnet Mask	Max Number of Host Possible	In Use	Network Name
36	220.0.1.64 (.65 **to** .126)	/26	62	✓	Research and Development LAN
18	220.0.1.160 (.161 **to** .190)	/27	30	✓	Sales and Marketing LAN
14	220.0.1.192 (.193 **to** .206)	/28	14	✓	Admin LAN
10	220.0.1.224 (.225 **to** .238)	/28	14	✓	Remote Sales LAN
2	220.0.1.252 (.253 **to** .254)	/30	2	✓	Branch-Main WAN
2	220.0.1.248 (.249 **to** .250)	/30	2	✓	Admin-Main WAN
2	220.0.1.240 (.241 **to** .242)	/30	2	✓	Admin WAN
-	220.0.1.0 (.1 **to** .62)	/26	62	✗	Unused
-	220.0.1.128 (.129 **to** .158)	/27	30	✗	Unused
3	220.0.1.208 (.209 **to** .215)	/29	6	✓	Management LAN
5	220.0.1.216 (.217 **to** .223)	/29	6	✓	Server LAN
-	220.0.1.244 (.245 **to** .246)	/30	2	✗	Unused

Other IP Address Ranges

Number of host addresses required	Network Address (Host Range)	Subnet Mask	Max Number of Host Possible	In Use	Network Name
2	200.200.100.0	/30	2	✓	Admin-ISP WAN
1	1.1.1.1	/32	1	✓	RSB1 OSPF RID
1	2.1.1.1	/32	1	✓	AR1 OSPF RID
1	2.2.2.2	/32	1	✓	AR2 OSPF RID
1	3.3.3.3	/32	1	✓	MBR1 OSPF RID

Router Interface Tables

This section includes the router interface tables of the routers used in XYZ's network topology, indicating the location, name, IP addressing, and interface details of each router.

Location: **Remote Sales Branch Office**

Router name: **RSB1**

Interface/ Sub-interface Type/Number	Description and purpose	DCE/ DTE	Speed / Clock rate	Network Name	Network Number	Interface IP Address	Subnet Mask
s0/0/0	Connect RSB1 to MBR1	DCE	128kbps	Branch-Main WAN	220.0.1.252	220.0.1.253	/30
Gig0/0	LAN for Remote Sales Office	-	1Gbps	Remote Sales LAN	220.0.1.224	220.0.1.225	/28
lo0	RSB1 OSPF RID	-	-	-	1.1.1.1	1.1.1.1	/32

Location: **Administration Building**

Router name: **AR1 (Edge)**

Interface/ Sub-interface Type/Number	Description and purpose	DCE/ DTE	Speed / Clock rate	Network Name	Network Number	Interface IP Address	Subnet Mask
Gig0/0	Connect AR1 to AR2	-	1 gbps	Admin WAN	220.0.1.240	220.0.1.241	/30
s0/0/1	Connect AR1 to MBR1	DTE	-	Admin - Main WAN	220.0.1.248	220.0.1.249	/30
s0/0/0	Connect AR1 to ISP	DCE	128 kbps	Admin - ISP WAN	200.200.100.0	200.200.100.1	/30
lo0	AR1 OSPF RID	-	-	-	2.1.1.1	2.1.1.1	/32

Location: **Administration Building**

Router name: **AR2**

Interface/ Sub-interface Type/Number	Description and purpose	DCE /DTE	Speed / Clock rate	Network Name	Network Number	Interface IP Address	Subnet Mask
Gig0/0	Connect AR2 to AR1	-	1gbps	Admin WAN	220.0.1.240	220.0.1.242	/30
Gig0/1	LAN for Admin Users	-	1gbps	Admin LAN	220.0.1.192	220.0.1.193	/28
lo0	AR2 OSPF RID	-	-	-	2.2.2.2	2.2.2.2	/32

Location: **Main Building**

Router name: **MBR1**

Interface/ Sub-interface Type/Number	Description and purpose	DCE /DTE	Speed / Clock rate	Network Name	Network Number	Interface IP Address	Subnet Mask
s0/0/0	Connect MBR1 to RSB1	DTE	-	Branch-Main WAN	220.0.1.252	220.0.1.254	/30
s0/0/1	Connect AR1 to MBR1	DCE	128kbps	Admin - Main WAN	220.0.1.248	220.0.1.250	/30
g0/0	LANs for Main Building	-	1gbps	-	-	-	-
g0/0.10	LAN for R & D users	-	-	Research and Development LAN	220.0.1.64	220.0.1.65	/26
g0/0.20	LAN for Sales & Marketing	-	-	Sales and Marketing LAN	220.0.1.160	220.0.1.161	/27
g0/0.30	LAN for Server	-	-	Server LAN	220.0.1.216	220.0.1.217	/29
g0/0.99	LAN for management	-	-	Management LAN	220.0.1.208	220.0.1.209	/29
lo0	MBR1 OSPF RID	-	-	-	3.3.3.3	3.3.3.3	/32

Switch Tables

This table includes the switches used in company XYZ's network topology, indicating the name, physical location, management IP address, encapsulation type (if required), and port information of each switch.

Location: **Remote Sales Branch Office**

Switch Name: RSS

Switch Management IP Address: **220.0.1.238**

Interface/Subinterface Type/Port/Number	Description and purpose	Speed	Duplex	Network Name	Network Number	Subnet Mask	VLAN	Switchport Type	Encapsulation (If needed)
g0/1	Connect to RSB1	1000 mpbs	Full	Remote Sales LAN	220.0.1.224	/28	1	Access	-
fa0/1-10	For remote sales LAN users	100 mpbs	Full	Remote Sales LAN	220.0.1.224	/28	1	Access	-

Location: **Administration Building**

Switch Name: **AIS**

Switch IP Address: **220.0.1.206**

Interface/Subinterface Type/Port/Number	Description and purpose	Speed	Duplex	Network Name	Network Number	Subnet Mask	VLAN	Switchport Type	Encapsulation (If needed)
g0/1	Connect to AIR	1000 mbps	Full	Admin LAN	220.0.1.192	/28	1	Access	-
fa0/1-14	For Admin building LAN users	100 mbps	Full	Admin LAN	220.0.1.192	/28	1	Access	-

Location: **Main Building Floor 1**

Switch Name: **MBS1**

Switch IP Address: **220.0.1.210**

Interface/Subinterface Type/Port/Number	Description and purpose	Speed	Duplex	Network Name	Network Number	Subnet Mask	VLAN	Switchport Type	Encapsulation
g0/1	Connect MBS1 to MBR	1000 mbps	full	-	-	-	10,20,30	Trunk	802.1q
fa0/1	Connect MBS2 to MBS1	100 mbps	full	Management LAN	220.0.1.208	/29	10,20,30	Trunk	802.1q
fa0/2	Connect MBS3 to MBS1	100 mbps	full	Management LAN	220.0.1.208	/29	10,20,30	Trunk	802.1q
fa0/3 - 7	Ports for Servers	100 mbps	full	Server LAN	220.0.1.216	/29	30	Access	-
fa0/8 - 15	Ports for R & D users on first floor	100 mbps /port	full	Research & Development LAN	220.0.1.64	/26	10	Access	-
fa0/16 - 21	Ports for Sales users on first floor	100 mbps /port	full	Sales & Marketing LAN	220.0.1.160	/27	20	Access	-

Location: **Main Building Floor 2**

Switch Name: **MBS2**

Switch IP Address: **220.0.1.211**

Interface/Subinterface Type/Port/Number	Description and purpose	Speed	Duplex	Network Name	Network Number	Subnet Mask	VLAN	Switchport Type	Encapsulation
fa0/24	Connect MBS1 to MBS2	100 mbps	full	Management LAN	220.0.1.208	/29	10,20,30	Trunk	802.1q
fa0/23	Connect MBS3 to MBS2	100 mbps	full	Management LAN	220.0.1.208	/29	10,20,30	Trunk	802.1q
f0/1-12	Ports for R&D users on second floor	100 mbps	full	Research & Development LAN	220.0.1.64	/26	10	Access	-
f0/13-20	Ports for Sales users on second floor	100 mbps	full	Sales & Marketing LAN	220.0.1.160	/27	20	Access	-

Location: **Main Building Floor 3**

Switch Name: **MBS3**

Switch IP Address: **220.0.1.212**

Interface/Subinterface Type/Port/Number	Description and purpose	Speed	Duplex	Network Name	Network Number	Subnet Mask	VLAN	Switchport Type	Encapsulation
fa0/24	Connect MBS1 to MBS3	100 mbps	full	Management LAN	220.0.1.208	/29	10,20 ,30	Trunk	802.1q
fa0/23	Connect MBS3 to MBS2	100 mbps	full	Management LAN	220.0.1.208	/29	10,20 ,30	Trunk	802.1q
fa0/1-16	Ports for R&D users on second floor	100 mbps	full	Research & Development LAN	220.0.1.64	/26	10	Access	-
fa0/17-20	Ports for Sales users on second floor	100 mbps	full	Sales & Marketing LAN	220.0.1.160	/27	20	Access	-

End devices Table (PC & Servers)

This section shows the list of host devices connected in XYZ's network topology. For demonstration purposes, all PC/workstation and server addresses will be statically assigned an IP address. In the actual topology, we will implement a DHCP server to dynamically allocate IP addresses to the PC/workstation, while servers will have static IP addresses.

LAN Name	PC or Server Name	IP Address	Subnet Mask	Gateway	Services Provided
Remote Sales LAN	RS_PC_1	220.0.1.226	/28	220.0.1.225	-
	RS_PC_2	220.0.1.227			
	RS_PC_3	220.0.1.228			
	RS_PC_4	220.0.1.229			
	RS_PC_5	220.0.1.230			
Admin LAN	A_PC_1	220.0.1.194	/28	220.0.1.193	-
	A_PC_2	220.0.1.195			

	A_PC_3	220.0.1.196			
	A_PC_4	220.0.1.197			
	A_PC_5	220.0.1.198			
	A_PC_6	220.0.1.199			
	A_PC_7	220.0.1.200			
Server LAN	HTTP Server	220.0.1.218	/29	220.0.1.217	Web Server
	TFTP Server	220.0.1.219	/29	220.0.1.217	TFTP Server
Research and Development LAN	RD_PC_1	220.0.1.66	/26	220.0.1.65	-
	RD_PC_2	220.0.1.67			
	RD_PC_3	220.0.1.68			
	RD_PC_4	220.0.1.69			
	RD_PC_5	220.0.1.70			
	RD_PC_6	220.0.1.71			
	RD_PC_7	220.0.1.72			
	RD_PC_8	220.0.1.73			
	RD_PC_9	220.0.1.74			
	RD_PC_10	220.0.1.75			
	RD_PC_11	220.0.1.76			
	RD_PC_12	220.0.1.77			
	RD_PC_13	220.0.1.78			
	RD_PC_14	220.0.1.79			

	RD_PC_15	220.0.1.80			
	RD_PC_16	220.0.1.81			
	RD_PC_17	220.0.1.82			
	RD_PC_18	220.0.1.83			
Sales and Marketing LAN	SM_PC_1	220.0.1.162	/27	220.0.1.161	-
	SM_PC_2	220.0.1.163			
	SM_PC_3	220.0.1.164			
	SM_PC_4	220.0.1.165			
	SM_PC_5	220.0.1.166			
	SM_PC_6	220.0.1.167			
	SM_PC_7	220.0.1.168			
	SM_PC_8	220.0.1.169			
	SM_PC_9	220.0.1.170			

Phase 2: Routing Protocols

In order to decide which routing protocol to use in company XYZ's network architecture, we have researched on multiple potential routing protocols to be used in the topology. By weighing the pros and cons of several common attributes between these protocols, we then determined the best protocol for use based on our defined metrics.

Routing Protocol Table

Routing Protocol	Property 1 (Resource)	Property 2 (Convergence)	Property 3 (Scalability)	Property 4 (Use of VLSM)	Property 5 (Cisco Proprietary)	Total
RIPv1	Low resource usage (+)	Low convergence Speed (-)	Low (-) Maximum hop count of 15	Classful (-)	Vendor Neutral (+)	2
RIPv2	Low resource usage (+)	Low convergence Speed (-)	Low (-) Maximum hop count of 15	Classless (+)	Vendor Neutral (+)	3
OSPF	High Resource Usage (-)	High convergence Speed (+)	High Scalability with OSPF areas (+)	Classless (+)	Vendor Neutral (+)	4
EIGRP	High Resource Usage (-)	High convergence Speed with Feasible Successor (+)	High Scalability (+)	Classless (+)	Cisco Proprietary (-)	3
BGP	Low Resource Usage (+)	Low convergence Speed (-)	High Scalability (+)	Classless (+)	Cisco Proprietary (-)	3

SSN's Recommendation for Routing Protocol in XYZ's network topology: **OSPF**

Based on our analysis and metric calculated, we have determined that OSPF would be the most appropriate routing protocol to be used in XYZ's topology based on its scalability, high convergence speed, and being a non-proprietary universal routing protocol which can be compatible with network devices from other brands if required by XYZ in the future.

Phase 3: OSPF

Routing Implementation

The XYZ Research Company has decided to use OSPF as the routing protocol after recommendations by us. We have determined XYZ Research Company requirements as per the following:

- OSPF process id 50
- All routers in a single area, Area 0
- Loopback address used on each router
- Loopback address used as the OSPF router ID
- There is one multi-access network in the Administration building and a DR would need to be elected
- Determine whether to use Router ID or interface priority to determine the DR in the building

As the topology only includes only a few routers between three buildings, we can design OSPF such that all routers can be included in the same area. Loopback addresses can be configured on each router for the router to determine its OSPF Router ID for identification purposes.

Deciding between using Router ID (RID) or highest loopback interface priority for DR Election

A decision needs to be made on which router and router interface will be the DR in the multi-access network (Admin-Main WAN). After some debate, we decided to use interface priority to determine DR as it ensures that a router with higher ospf interface priority value would indefinitely become the DR. Using router ID to determine DR may change the DR and BDR in the event that the loopback interfaces of the routers change. This can be avoided by configuring interface priority and hence we used that to determine the DR in our topology.

Deciding which router will be the DR

After deciding on whether to use router ID or interface priority to determine DR, we chose router **AR1 in the Administration Building** as the DR. We have configured router **AR1's g0/0** interface to have a **priority value of 69**, which is higher than AR2's g0/0 interface so AR1 will always be elected as a DR. We made AR1 the DR as most of the OSPF networks shared by AR1 originates from AR1 onwards to the rest of the topology, so in the case of future expansion of routers in the multi-access network between AR1 and AR2, we would want AR1 to continue to be the primary DR as it is facing towards the rest of the topology. As the DR collects the routes learnt by the other routers in the multi access network, it in theory would also decrease the bandwidth used to send OSPF updates and routes if AR1 is the DR. Hence, we have strategically chosen AR1 to be the DR.

Configuring OSPF DR in XYZ's topology

As DR election only happens on routers connected with multi-access links (such as the Admin-Main WAN), DR elections would not need to be elected on serial connections between the buildings and the remote branch office. As such, configurations relating to DR would be done only on the Admin-Main WAN link.

OSPF Configuration Additional Notes

A static default route is configured on router AR1 acting as the edge router to the ISP and internet, and OSPF is configured on AR1 to propagate this default route to the rest of the routers in Area 0.

Passive interfaces are also configured on the router's ports connecting to any LAN with end devices, which decreases unnecessary OSPF traffic on the LANs.

With phase 2 and 3 completed, we can begin to configure OSPF on XYZ's routers in the topology with the information.

Phase 4: VLANS

VLAN, STP, Switches Implementation

As the main building consists of different departments with their own LANs and end devices across multiple floors, VLANs would need to be configured on multiple switches between the different floors of XYZ's main building.

Company XYZ has requested for us to plan out the configuration on their switches (VTP, STP, and more) in their Main Building as per their requirements, as listed below:

- There are 18 Research and Development employees
- There are 9 Sales and Marketing employees
- 8 Research personnel and 2 Sales personnel are on Floor 3
- 6 Research personnel and 4 Sales personnel are on Floor 2
- 4 Research personnel and 3 Sales personnel are on Floor 1
- Floor 1 has a Server LAN that has one server currently with the maximum capacity of 5 servers.
- There are separate rooms on each floor for the workgroups
- There is sufficient space in each room to accommodate growth
- The 100% predicted growth will occur uniformly per floor
- Switches in the main building are connected in a loop so that if one switch fails an alternate route is formed.
- There will be one switch per floor shared among the different subnets
- Spanning tree root (for each VLAN) is to be placed in an optimal position.

The VLAN assignments for each LAN in the Main Building are as follows:

VLAN Table

VLAN Number	LAN Name
10	Research & Development LAN
20	Sales & Marketing LAN
30	Server LAN
60	Unused LAN (For unused ports)
99	Management LAN

The purpose of VLAN 60 is to store ports that are not currently used by the switches for better security practices, and VLAN 99 is used to remotely manage the switches in the main building.

The table below shows the switch names that will be used in XYZ's topology, along with other information such as the hardware (model name and number of ports), location, IP addressing information, VTP, and STP statuses.

Detailed Switch Table A

Switch Name	Model	# of Ports	Location	IP Address	Gateway	Management VLAN	VTP Mode	VTP Domain	STP Root
MBS1	2960-24TT	24	Main Building Floor 1	220.0.1.210	220.0.1.209	99	Server	MBS1	VLAN 30
MBS2	2960-24TT	24	Main Building Floor 2	220.0.1.211	220.0.1.209	99	Client	MBS1	VLAN 20
MBS3	2960-24TT	24	Main Building Floor 3	220.0.1.212	220.0.1.209	99	Client	MBS1	VLAN 10
AIS	2960-24TT	24	Administration Building	220.0.1.206	220.0.1.193	1	-	-	-
RSS	2960-24TT	24	Branch Office	220.0.1.238	220.0.1.225	1	-	-	-

Detailed Switch Table B

Switch Name	# of Ports	Access Ports	Trunk Ports
MBS1	24	Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/2	Fa0/1, Fa0/2, Gig0/1
MBS2	24	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22	Fa0/23, Fa0/24, Gig0/1, Gig0/2
MBS3	24	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22	Fa0/23, Fa0/24, Gig0/1, Gig0/2

Switch Ports Configuration

All switch-to-switch ports have been set to trunking, VLANS 10,20,30,99 have been propagated across the switches via VTP configuration. The VLAN assignment of access ports for each switch can be found in the Switch Tables in phase 1, which follows the requirements.

In addition, we have configured Portfast and BPDU-Guard on all access ports on all switches. BPDU-Guard will protect against rogue switches from being connected to the main building switches. Portfast will ensure that end devices connected to the switch can immediately connect to the network.

STP Root Bridge Decision

Regarding the selection of STP Root Bridge(s) for the different VLANs, we made the choices based on the number of hosts connected to each VLAN on the switches. For example, there are 8 Research & Development (R&D) staffers connected to MBS3, as compared to only 4 R&D staffers connected to MBS1. By determining the root bridge of a particular VLAN by seeing if that switch has the most hosts belonging to that VLAN, each of the VLANs (10, 20, and 30) are root bridges in different switches in the main building. This will load balance spanning-tree traffic and overall improve the bandwidth efficiency in the main building LAN.

In addition to the STP Root bridges that we have decided for each VLAN, we have also decided on secondary root bridges for the following spanning-tree processes:

Spanning-tree bridge table summary

Spanning-tree Process	Primary Root Bridge	Secondary Root Bridge
VLAN 10	MBS3	MBS2
VLAN 20	MBS2	MBS1
VLAN 30	MBS1	MBS3

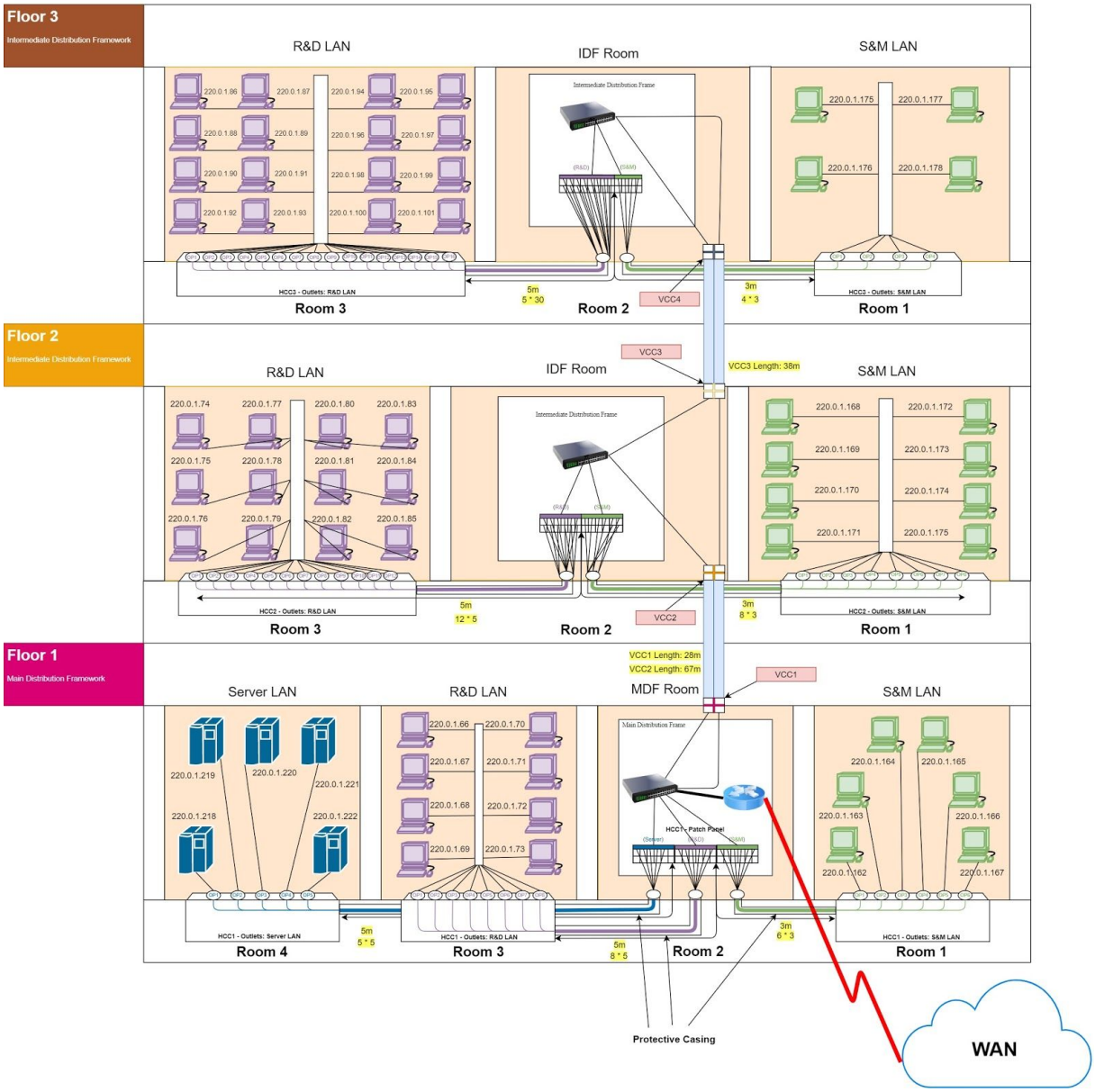
With spanning-tree configured on MBS1, MBS2, and MBS3, it provides redundancy if one link between two of those switches goes down, and we have optimized the bandwidth usage within the switches in the main building by optimally placing the root bridges for the main building VLANs.

Phase 5: Physical Layer Design

Physical Layer Design

As part of the requirements, we have drafted several sample physical layer designs that include the growth projections of the company. To note, these designs are only for the company's Main Building, and are intended to provide visualization.

Main Building Mock-up Diagram (Diagram not drawn to scale)



The Main Building Mock-up Diagram was designed to be a physical representation of the building from an X-rayed front view. Based on the provided specifications from the XYZ Company, we identified the number of floors and rooms which would comprise the building. Similarly, we also identified the number of personnel in each floor and room, and was thus able to calculate the number of equipment needed to establish the network architecture for the Main Building.

As part of our mock-up diagram, we attempted to illustrate the backbone cabling methods which would be used as part of the network. Notably, we used Cat 8.2 Solid Core Ethernet cables for the backbone cabling (between floors and rooms), as these are intended to be long term installations. We also made sure to avoid bending the cables as much as possible in order to maintain it in the best condition. Contrastingly, we used Cat 8.2 Stranded Ethernet cables for the connections between end devices and the floor outlets (termination points), as well as for connections between network devices (switch) and patch panels. The pinout for all cables involved in Horizontal Cross-Cabling and Vertical Cross-Cabling is straight-through. Cable ties are used for cable management, especially for the Horizontal Cross-Cabling and Vertical Cross-Cabling, to form Ethernet trunk cables.

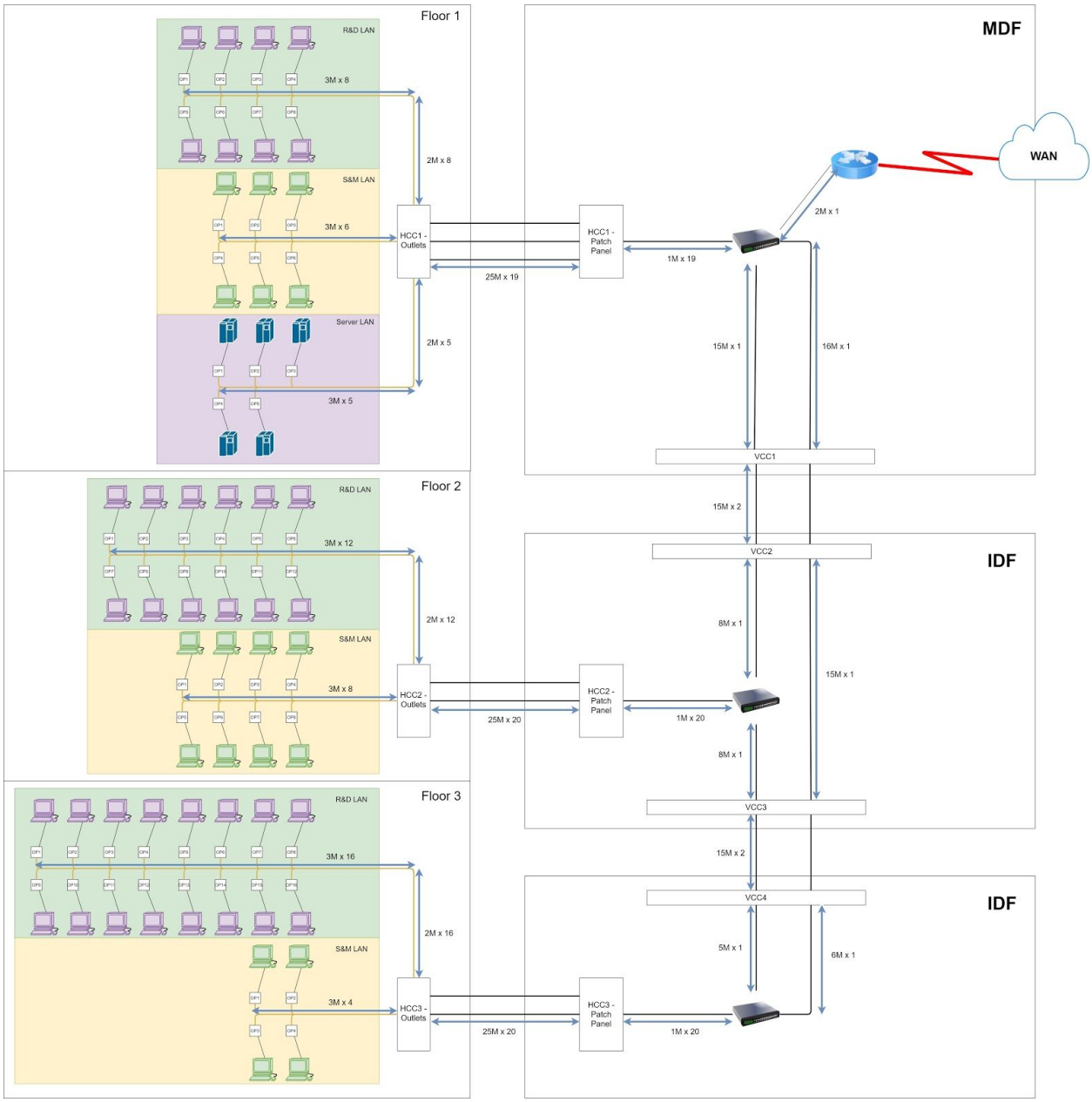
For Horizontal Cross-Cabling between the various rooms on a single floor, we came up with the idea of implementing raised flooring in the building. This would allow the RJ45 keystone jacks (for cabling termination) to be mounted under the raised flooring (as floor outlets), hidden and protected by floor panels. The trunk cabling under the raised flooring would run through PVC pipes, in order to protect them from damage. The PVC pipes (and cabling within) would all converge below the designated room containing the distribution framework. The other end of the trunk cabling would exit from these pipes and connect to the patch panel serving as a horizontal cross-connect. Single ethernet cables would then connect the patch panel to the switch (one individual cable per port, instead of a trunk cable).

For Vertical Cross-Cabling between the three floors, we utilise individual stranded Ethernet cables to connect the switches to wall-mounted patch panels. Solid Core Ethernet cables that are trunked together (using cable tie) then connect the patch panels between the different floors. These connections will run through PVC pipes installed as part of the building's architecture, in order to protect and facilitate the Vertical Cross-Cabling.

For the connections between end devices and floor outlets, stranded Ethernet cables will be most suitable for the task, as they are highly flexible as compared to their solid counterparts. There have been enough outlets installed in each room to account for 100% growth of the company.

Main Building Abstract Diagram

(Floors are in reverse order. Diagram not drawn to scale)



The Main Building Abstract Diagram was designed to be a logical representation of the physical layer of the network. From the provided specifications, we determined the lengths needed for each cable run between the different devices comprising the network architecture.

Similarly to our Mock-up Diagram, we included the necessary labeling of the floors, LANs, wall/floor outlets, Horizontal Cross-Connects, Vertical Cross-Connects, and network devices. The abstract diagram is intended to provide a simpler overview of the physical network architecture.

Details of Main Building Cabling:

- Ethernet Category used: Category 8.2 (throughout the building)
- Pinout type: Straight-through (for whole of network)

- Total length of Ethernet Cabling: 1928m
    - Total length of Solid Core Ethernet Cabling: 1475m
    - Total length of Stranded Ethernet Cabling: 393m

    - Total length of Cabling for Floor 1: 610m
        - Length of Solid Core Ethernet Cabling: 475m
        - Length of Stranded Ethernet Cabling: 135m

    - Total length of Cabling for Floor 2: 635m
        - Length of Solid Core Ethernet Cabling: 500m
        - Length of Stranded Ethernet Cabling: 135m

    - Total length of Cabling for Floor 3: 623m
        - Length of Solid Core Ethernet Cabling: 500m
        - Length of Stranded Ethernet Cabling: 123m

    - Length of Cabling between VCC1 and VCC2: 30m
    - Length of Cabling between VCC3 and VCC4: 30m

- HCC Backbone Cabling connects Outlet Port No. to matching Patch Panel Port No.
    - Example: HCC1 - Outlets: RD-OP5 connects to HCC1 - Patch Panel: RD-PP5
- VCC Backbone Cabling connects via matching Patch Panel Port No.
    - Example: VCC1: Floor1xFloor2-PP3 connects to VCC2: Floor2xFloor1-PP3

- Network Distribution Frameworks are located in Room 2 for each floor.

Solid-conductor cable is designed for backbone and horizontal cable runs. As such, it should be used for runs between two wiring closets or from the wiring closet to a network outlet.The cable should not be bent, flexed, or twisted repeatedly. Its attenuation is lower than that of stranded-conductor cable, making it the optimal choice for long-term installations.

Stranded cable is for use in shorter runs between network interface cards (NICs) and network outlets, or between concentrators and patch panels, hubs, and other rack-mounted equipment. Stranded-conductor cable is much more flexible than solid-core cable. However, attenuation is higher in stranded-conductor cable, so the total length of stranded cable in your system should be kept to a minimum to reduce signal degradation.

Labeling Convention (for HCC - Patch Panels)

HCC1 - Patch Panel (Floor 1):

HCC1 - Patch Panel

(Server)						(R&D)						(S&M)					
S-PP1	S-PP2	S-PP3	S-PP4	S-PP5	S-PP6	RD-PP1	RD-PP2	RD-PP3	RD-PP4	RD-PP5	RD-PP6	SM-PP1	SM-PP2	SM-PP3	SM-PP4	SM-PP5	SM-PP6
S-PP7	S-PP8	S-PP9	S-PP10	S-PP11	S-PP12	RD-PP7	RD-PP8	RD-PP9	RD-PP10	RD-PP11	RD-PP12	SM-PP7	SM-PP8	SM-PP9	SM-PP10	SM-PP11	SM-PP12

HCC2 - Patch Panel (Floor 2):

HCC2 - Patch Panel

(R&D)						(S&M)					
RD-PP1	RD-PP2	RD-PP3	RD-PP4	RD-PP5	RD-PP6	SM-PP1	SM-PP2	SM-PP3	SM-PP4	SM-PP5	SM-PP6
RD-PP7	RD-PP8	RD-PP9	RD-PP10	RD-PP11	RD-PP12	SM-PP7	SM-PP8	SM-PP9	SM-PP10	SM-PP11	SM-PP12

HCC3 - Patch Panel (Floor 3):

HCC3 - Patch Panel

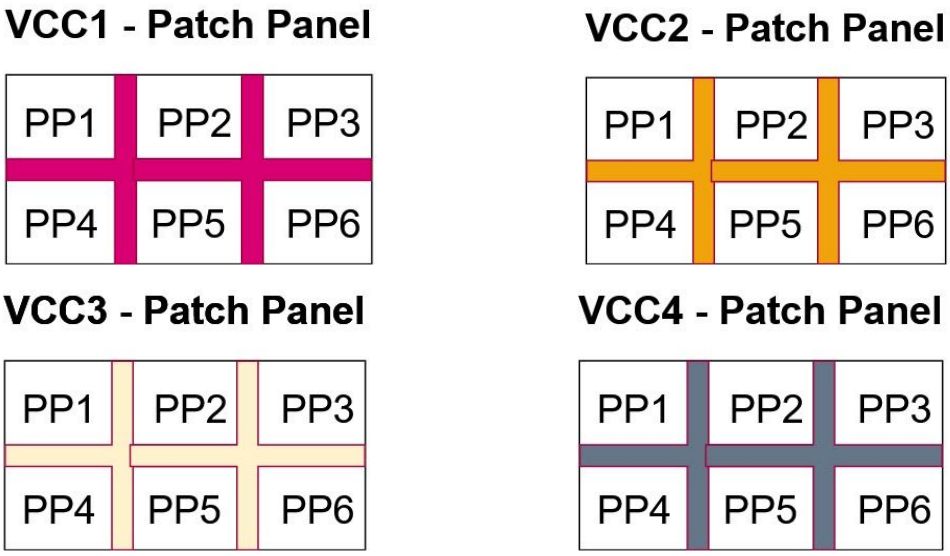
(R&D)								(S&M)			
RD-PP1	RD-PP2	RD-PP3	RD-PP4	RD-PP5	RD-PP6	RD-PP7	RD-PP8	SM-PP1	SM-PP2	SM-PP3	SM-PP4
RD-PP9	RD-PP10	RD-PP11	RD-PP12	RD-PP13	RD-PP14	RD-PP15	RD-PP16	SM-PP5	SM-PP6	SM-PP7	SM-PP8

For the HCC - Patch Panel labeling, the patch panel ports are separated into different LAN groups. These groups are the “Server” LAN, “Research and Development” LAN, and “Sales and Marketing” LAN. As part of the labeling scheme, they are denominated as “S”, “R&D” and “S&M”

respectively. The “-PP” associated with each port is to identify it as a patch panel port, while the numbering behind is to mark each port uniquely within its own LAN group.

To identify between patch panel ports on different floors, the floors themselves will be used as part of the labeling. A sample label would be “HCC1 - Patch Panel: RD-PP5”, whereby the “1” refers to the floor which the patch panel is on. As part of the Main Building Mock-up diagram, all patch panels are located in Room 2, regardless of floor.

Labeling Convention (for VCCs)

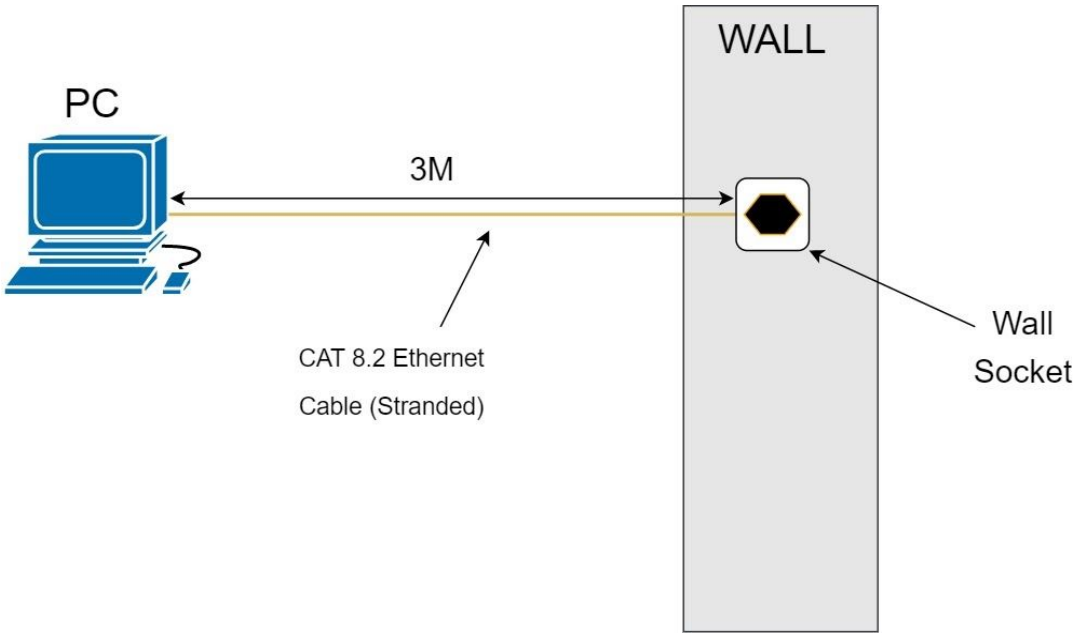


For the VCC labeling, it is quite simple in nature due to only being few in number. The patch panel ports are labeled “1” through “6” as a unique marker for the ports. Similarly, the VCCs themselves are labeled “1” through “4” as a unique marker among themselves.

- VCC1 resides on Floor 1, and facilitates the connection between Switch 1 and VCC2
- VCC2 resides on Floor 2, and facilitates the connection between Switch 2 and VCC1, as well as VCC3 and VCC1
- VCC3 resides on Floor 2, and facilitates the connection between Switch 2 and VCC4, as well as VCC2 and VCC4
- VCC4 resides on Floor 3, and facilitates the connection between Switch 3 and VCC3

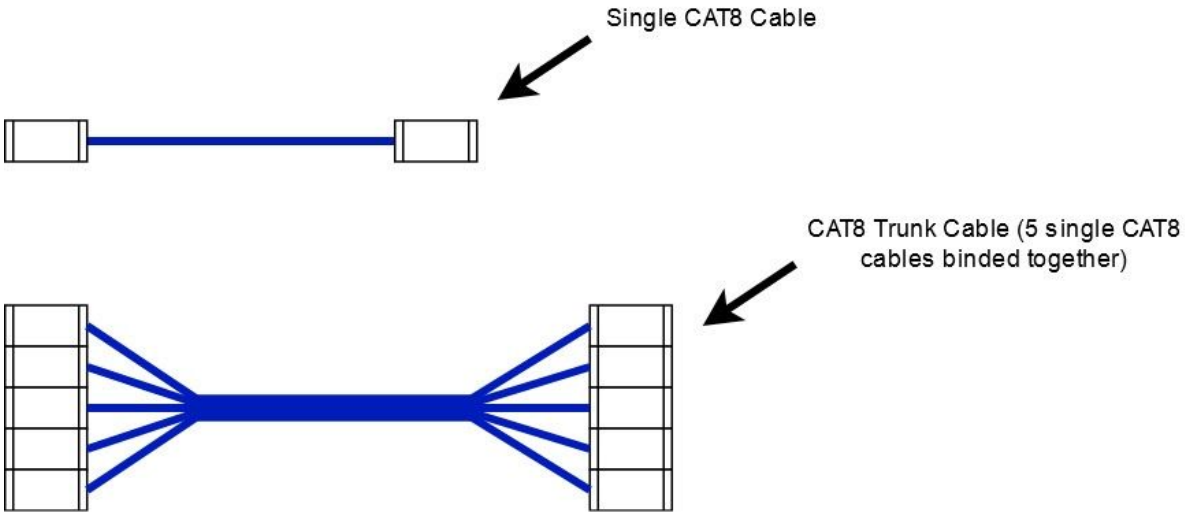
To assist in identification, the VCC labeling includes the floors which they connect. A sample label would be “VCC1: Floor1xFloor2-PP3”, whereby the “1” refers to the floor which the patch panel is on. As part of the Main Building Mock-up diagram, all VCCs are located in Room 2, regardless of floor.

Sample PC - Outlet Diagram:



The Sample PC - Outlet Diagram illustrates a typical connection between a host and a termination outlet. For our sample version of XYZ's building layout, we estimated 3 meters of cable to be sufficient to connect each host to a nearby outlet in the same room. The cables used for host-outlet connections will be stranded Cat 8.2 Ethernet cables, which meets the company's requirement to use the highest category cable throughout the LANs. The selection of stranded over solid for the cable type is due to the former's flexibility being suitable for connections to host devices which may move around.

Sample Trunk Cable Diagram:



The Sample Trunk Cable Diagram illustrates how a trunk cable is created for use within XYZ's network. While commonly-used trunk cables tend to be a bound of individual copper cables which are factory terminated, we have opted to create the trunk cables instead of sourcing them from factories. The reason for this is due to Category 8 being a relatively new Ethernet standard in the market. Very few companies sell pre-terminated Cat 8 trunk cables; They will have to be custom-ordered at exorbitant prices. Due to the lack of a definite price tag and unproven results, we have decided to create the trunk cables for use within XYZ's network by binding multiple Cat 8 cables with heavy-duty cable ties. Due to the small scale of XYZ's Main Building, the typical issues of fault and material waste associated with self-created trunk cables are little to none.

Physical Layer Equipment

Having designed the physical layer diagrams to represent the network, we then sourced for the various equipment necessary for establishing it. To ensure that cost prices were not over-inflated, we made sure to source the pricing of the equipment from various online retailers. We then took the average price calculated from the various sources. The list of retailers sourced for the various equipment prices can be found in Appendix C: Pricing References.

Main Building Cable Parts

Equipment Type	Model No.	Qty	Cost (SGD)
Ethernet Cable (Solid core)	CAT8.2	10 Spool (1 spool = 150m, 10 spool needed - 1475m)	$9,500
Ethernet Cable (Stranded)	CAT8	2 Spool (1 spool = 305m, 2 spool needed - 393m)	$1,787
RJ45 Jacket	CAT8.2	124 (1 box = 100pcs, 2 boxes needed)	$47.60
RJ45 Wall Socket	CAT8.2	59	$2,082.70
PVC Pipe	Georg Fischer	800 pipes, 2m long x 60mm OD, 4.5mm Wall Thickness	$36,410
Cable Tie	Heavy Duty 26 Inch	91 (1 packet = 50pcs, 2 packets needed)	$44 Total: $49,871.30

Due to the various reasons mentioned previously, we have decided to create, install and terminate the cabling ourselves. As such, we have opted to purchase spools of cables, RJ45

jackets and wall sockets in order to create the necessary cables. Cable ties will also be used to form trunk cables, and to ensure proper cable management during installation.

Main Building Floor 1, Room 2, MDF

Equipment Type	Model No.	Qty	No. /Type Ports	Description/ Function	Cost (SGD)
Router	2901	1	2 integrated 10/100/1000 Ethernet ports (Gig)	Forwards data packets between computer networks.	$1,900
Serial Network Module	HWIC-2T	1	2 serial interfaces	Provide serial interfaces to the router	$200
Switch	2960-24TT	1	24 integrated 10/100 Ethernet ports (Fa), 2 integrated 100/1000 Ethernet ports (Gig)	Connects to other network devices and end devices.	$1,400
24 ports Patch Panel	WC351088	1	24 Ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting, and testing circuits in a convenient, flexible manner	$475
6 ports Patch Panel	Yankok [6 Port Patch Panel]	1	6 ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting, and testing circuits in a convenient, flexible manner	$15
Rack	QFR1924-45-BK	1	-	Rack to store/hold the network equipment and patch panel	$445 Total: $4,435

Main Building Floor 2, Room2, IDF

Equipment Type	Model No.	Qty	No. /Type Ports	Description/ Function	Cost
Switch	2960-24TT	1	24 integrated 10/100 Ethernet ports (Fa), 2 integrated 100/1000 Ethernet ports (Gig)	Connects to other network devices and end devices.	$1,400
24 ports Patch	WC351088	1	24 Ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting,	$475

Panel				and testing circuits in a convenient, flexible manner	
6 ports Patch Panel	Yankok [6 Port Patch Panel]	2	6 ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting, and testing circuits in a convenient, flexible manner	$30
Rack	QFR1924-45-BK	1	-	Rack to store/hold the network equipment and patch panel	$445
					Total: $2,350

Main Building Floor 3, Room 2, IDF

Equipment Type	Model No.	Qty	No. /Type Ports	Description/ Function	Cost
Switch	2960-24TT	1	24 integrated 10/100 Ethernet ports (Fa), 2 integrated 100/1000 Ethernet ports (Gig)	Connects to other network devices and end devices	$1,400
24 ports Patch Panel	WC351088	1	24 Ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting, and testing circuits in a convenient, flexible manner	$475
6 ports Patch Panel	Yankok [6 Port Patch Panel]	1	6 ports, RJ45 and rear 110-type terminations.	Connects routing circuits for monitoring, interconnecting, and testing circuits in a convenient, flexible manner	$15
Rack	QFR1924-45-BK	1	-	Rack to store/hold the network equipment and patch panel	$445
					Total: $2,335

Phase 6: Test Network

By this phase, SSN has configured and installed the physical equipment in their sites and have configured all end devices and servers based on the equipment bought and provided in phase 5. In this phase, we will be testing the network connectivity based on the configured network equipments in the following tests requested by XYZ:

- Demonstrate routing between Main Building VLANS
- Demonstrate access of all hosts to each other and http service on the server machine
- Demonstrate behavior of the internetwork when a single trunk link fails

Testing & Verification of Network Devices

Demonstrate routing between Main Building VLANs

To demonstrate the routing between VLANs in the Main building, three tests can be performed between VLANs 10, 20, and 30.

1) Test the ping connectivity between hosts in VLAN 10 to hosts in VLAN 20
2) Test web access to the web server in VLAN 30 from hosts in VLAN 10
3) Test web access to the web server in VLAN 30 from hosts in VLAN 20

Table Demonstrating Routing between Main Building VLANS

From VLAN/SW	To VLAN/SW	Protocol	Hosts	Pass / Fail
10	20	ICMP	Hosts in Research & Development LAN and Sales & Marketing LAN	Pass
10	30	HTTP	Hosts in Research & Development LAN and a HTTP Server	Pass
20	30	HTTP	Hosts in Sales & Marketing LAN and a HTTP Server	Pass

More in-depth results of the ping and web connectivity will be shown in the next section.

Demonstrate access of all hosts to each other and http service on the server machine

In order to demonstrate these tests, we have randomly selected two hosts from each of XYZ's LAN and tested if they are reachable to each other via ping commands. We also tested these hosts to see if they can access the website on XYZ's web server in the Main building.

Table Demonstrating Access of all Hosts to Each Other and HTTP Service on the Server Machine

From Host	To Host	Pass / Fail
RD_PC_1 (220.0.1.66)	SM_PC_1 (220.0.1.162)	Pass
	RS_PC_1 (220.0.1.226)	Pass
	HTTP Server (220.0.1.218)	Pass
	A_PC_1 (220.0.1.194)	Pass
SM_PC_2 (220.0.1.163)	RD_PC_2 (220.0.1.67)	Pass
	RS_PC_2 (220.0.1.227)	Pass
	HTTP Server (220.0.1.218)	Pass
	A_PC_2 (220.0.1.195)	Pass
RS_PC_3 (220.0.1.228)	RD_PC_3 (220.0.1.68)	Pass
	SM_PC_3 (220.0.1.164)	Pass
	HTTP Server (220.0.1.218)	Pass
	A_PC_3 (220.0.1.196)	Pass
A_PC_4 (220.0.1.197)	RD_PC_4 (220.0.1.69)	Pass
	RS_PC_4 (220.0.1.229)	Pass
	HTTP Server (220.0.1.218)	Pass
	SM_PC_4 (220.0.1.165)	Pass
HTTP Server (220.0.1.218)	RD_PC_5 (220.0.1.70)	Pass
	RS_PC_5 (220.0.1.230)	Pass
	SM_PC_5 (220.0.1.166)	Pass
	A_PC_5 (220.0.1.198)	Pass

The results show that most hosts from different sites have access to each other and have access to http service on the server machine. The output of the tests in the above table can be found in Appendix A.

Demonstrate behavior of the internetwork when a single trunk link fails

In this section, we will demonstrate how the different instances of spanning tree converge in the event that each of the three links between MBS1, MBS2, and MBS3 fails. To test for a spanning-tree instance, we will conduct a ping test from an end device to another end device connected to a different switch on the same VLAN (i.e. ping between floors of the main building of the same department). Our network engineers have used various tools to determine the path of the ping traffic and how it changes when a trunk is manually disabled for testing purposes.

Table Demonstrating Behavior of the Internetwork When a Single Trunk Link Fails

From Host	To Host	Trunk up/down	Route	Pass / Fail
RD_PC_11 (VLAN 10, MBS3)	RD_PC_1 (VLAN 10, MBS1)	Up	RD_PC_11 → MBS3 → MBS1 → RD_PC1	Pass
		Down (Trunk Between MBS1 & MBS3)	RD_PC_11 → MBS3 → MBS2 → MBS1 → RD_PC1	Pass
SM_PC_4 (VLAN 20, MBS2)	SM_PC_8 (VLAN 20, MBS3)	Up	SM_PC_4 → MBS2 → MBS3 → SM_PC_8	Pass
		Down (Trunk between MBS2 & MBS3)	SM_PC_4 → MBS2 → MBS1 → MBS3 → SM_PC_8	Pass

Based on the results, our engineers have determined that the spanning tree processes are able to converge properly in the event of a trunk link failure, ensuring network reliability and redundancy.

With that, we have successfully concluded that the network has passed all tests, and is viable for deployment in a production capacity.

Future Recommendations for Network Growth

While company XYZ's topology is sufficient to meet their network requirements, our network engineers at SNN Consultant Group have identified some potential areas of improvements for future network growth and design:

- Currently, there are no ACLs placed on the subinterface (g0/0.30) of the Server LAN in the main building router (MBR1), so a ACL could be placed to only permit web traffic for their web server, and other unused protocols can be blocked by the ACL for security purposes.
- In the links between switches and routers in the main building (involving MBS1, MBS2, and MBS3), etherchannel can be implemented to increase the redundancy of the LAN of several links going down concurrently, and increase the bandwidth available for greater traffic volume.
- Similarly, multilink can be configured on router-to-router links between XYZ's sites to increase network redundancy in the WAN.
- Hot Standby Routing Protocol (HSRP) can be configured in the main building to increase the redundancy of the Main Building's LAN in case the link between the edge switch (MBS1) and router (MBR1) goes down. However, an additional router is required in the main building for this.
- If there are not enough ports and IP addresses for XYZ's HTTP server in their Main Building Server LAN, HTTP Servers can be implemented in a Cloud Computing environment (AWS/Azure).
- Intrusion Detection Systems and Intrusion Prevention Systems can be implemented in each of XYZ's sites for additional network security on top of router ACLs.
- Link Aggregation Control Protocol (LACP) increases bandwidth by trunking ports and includes redundancy, which servers with dual-ethernet ports can benefit from such as XYZ's HTTP server.

These network recommendations could be implemented in future network plannings, which will greatly increase the security, redundancy, and reliability of XYZ's network.

Phase 7: Final Demonstration

In this phase, we will conduct a final demonstration of the complete company internetwork, including testing internet connectivity for all end devices in all of XYZ's sites.

The table below demonstrates two hosts in each of XYZ's LANs and tests their network connectivity with the internet server (150.13.2.1).

Table of Test Summary Results

From Host	To Host	Path Tested	Route Taken	Pass / Fail
RD_PC_1 (220.0.1.66)	Internet Server (150.13.2.1)	MBS1 to ISP	RD_PC_1 → MBS1 → MBR1 → AR1 → ISP	Pass
RD_PC_8 (220.0.1.73)		MBS2 to ISP	RD_PC_8 → MBS2 → MBS3 → MBS1 → MBR1 → AR1 → ISP	Pass
RD_PC_11 (220.0.1.76)		MBS3 to ISP	RD_PC_11 → MBS3 → MBS1 → MBR1 → AR1 → ISP	Pass
SM_PC_1 (220.0.1.162)		MBS1 to ISP	SM_PC_1 → MBS1 → MBR1 → AR1 → ISP	Pass
SM_PC_4 (220.0.1.165)		MBS2 to ISP	SM_PC_4 → MBS2 → MBS1 → MBR1 → AR1 → ISP	Pass
SM_PC_8 (220.0.1.169)		MBS3 to ISP	SM_PC_8 → MBS3 → MBS2 → MBS1 → MBR1 → AR1 → ISP	Pass
HTTP Server (220.0.1.218)		MBS1 to ISP	HTTP Server → MBS1 → MBR1 → AR1 → ISP	Pass
RS_PC_1 (220.0.1.226)		RSS to ISP	RS_PC_1 → RSS → RSB1 →	Pass

			MBR1 → AR1 → ISP	
A_PC_1 (220.0.1.194)		AIS to ISP	A_PC_1 → AIS → AR2 → AR1(Edge) → ISP	Pass

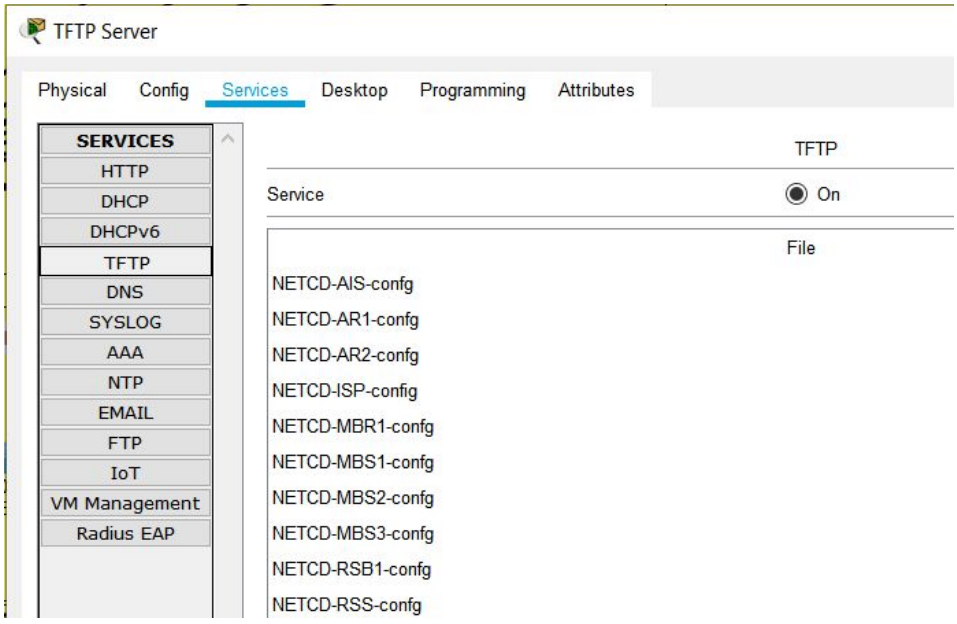
Note: For demonstration purposes, the internet server is configured on the ISP router on loopback interface 0

TFTP Configuration

We have decided to store our network testing results in a TFTP server in XYZ's main building, including the running configuration, routing tables, and ping tests. The output of the ping test between hosts and web server access can be found in Appendix A.

In addition to the ping tests, the running configuration of each of the routers and switches have been saved to that TFTP server. They can be identified with the prefix of "NETCD", followed by the network device name.

TFTP Server Files



The routing tables of the routers used in XYZ's topology are also copied over to TFTP server, and the outputs of the routing table can be found on Appendix B.

Note: Cisco Packet Tracer only supports copying the running-configuration of network devices to a TFTP server only, and does not support the functionality of copying output of ping statistics and routing table to the TFTP server.

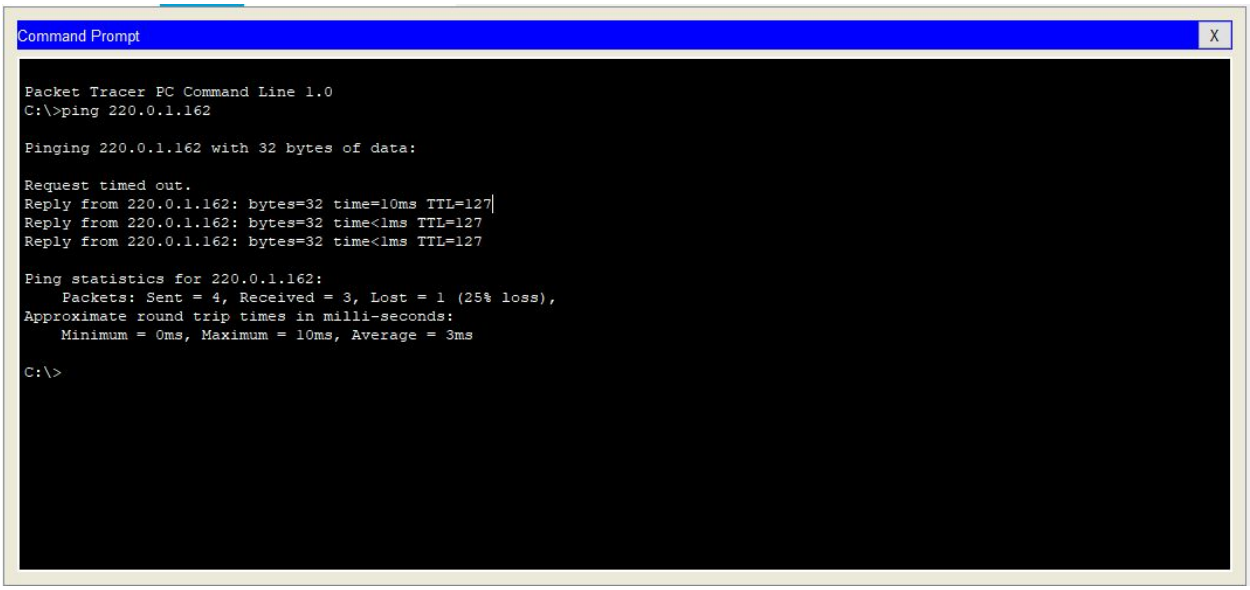
With phase 6 and phase 7 done, we have identified all aspects of the network to be functioning as intended.

Conclusion

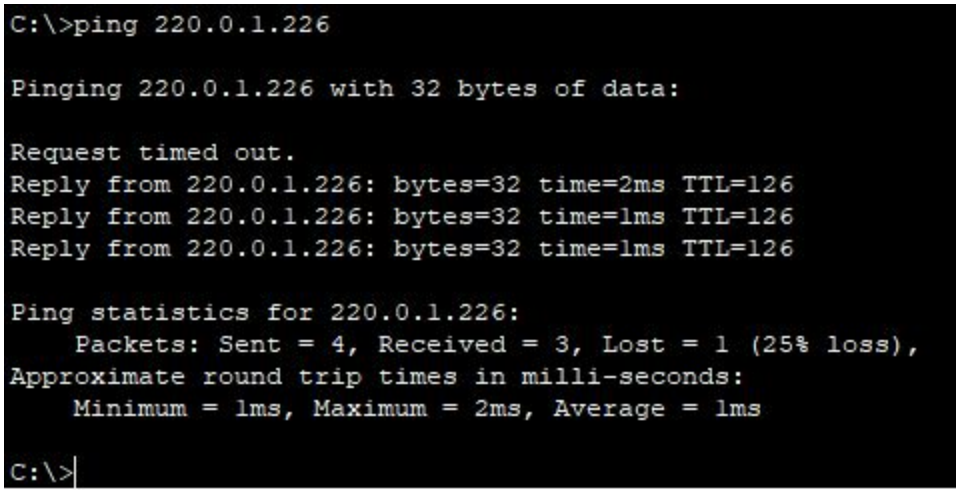
We would like to wrap up our report with a brief summary of the overall project. In this report, we have covered the network requirements, routing and switching protocols, physical layer design, necessary equipment and troubleshooting of the network. A list of recommendations has also been compiled within the document, to provide an idea of the potential areas of improvement within the network. We have also subjected the network architecture to a series of extensive tests, which have all successfully cleared. As part of our demonstration of the designed network, we have prepared a Packet Tracer simulation in order to present the completed network topology.

Appendix A: Testing Results

RD_PC_1(220.0.1.66) to SM_PC_1 (220.0.1.162)



RD_PC_1(220.0.1.66) To RS_PC_1 (220.0.1.226)



RD_PC_1(220.0.1.66) to HTTP Server (220.0.1.218)

```
C:\>ping 220.0.1.218

Pinging 220.0.1.218 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.218: bytes=32 time=1ms TTL=127
Reply from 220.0.1.218: bytes=32 time=10ms TTL=127
Reply from 220.0.1.218: bytes=32 time<1ms TTL=127

Ping statistics for 220.0.1.218:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

RD_PC_1(220.0.1.66) to A_PC_1 (220.0.1.194)

```
C:\>ping 220.0.1.194

Pinging 220.0.1.194 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.194: bytes=32 time=1ms TTL=125
Reply from 220.0.1.194: bytes=32 time=1ms TTL=125
Reply from 220.0.1.194: bytes=32 time=16ms TTL=125

Ping statistics for 220.0.1.194:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 6ms

C:\>
```

SM_PC_2 (220.0.1.163) to RD_PC_2 (220.0.1.67)

```
C:\>ping 220.0.1.67

Pinging 220.0.1.67 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.67: bytes=32 time<1ms TTL=127
Reply from 220.0.1.67: bytes=32 time<1ms TTL=127
Reply from 220.0.1.67: bytes=32 time=1ms TTL=127

Ping statistics for 220.0.1.67:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

SM_PC_2 (220.0.1.163) to RS_PC_2 (220.0.1.227)

```
C:\>ping 220.0.1.227

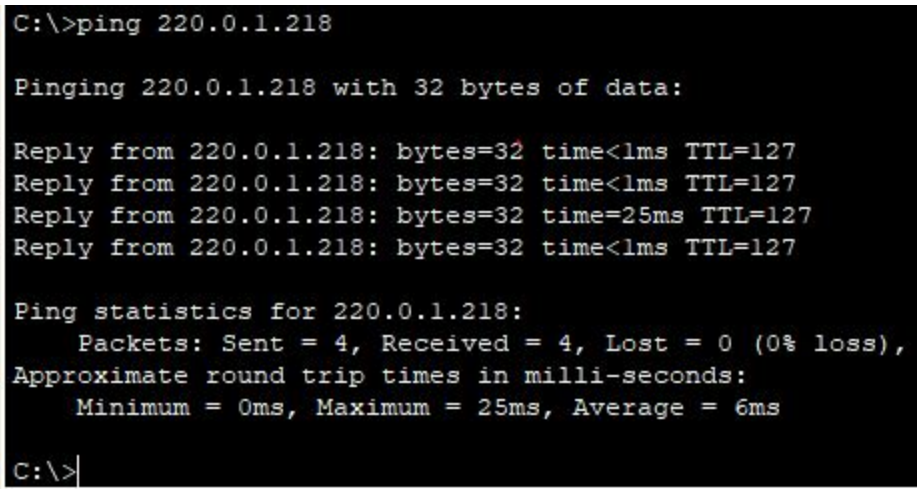
Pinging 220.0.1.227 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.227: bytes=32 time=12ms TTL=126
Reply from 220.0.1.227: bytes=32 time=12ms TTL=126
Reply from 220.0.1.227: bytes=32 time=1ms TTL=126

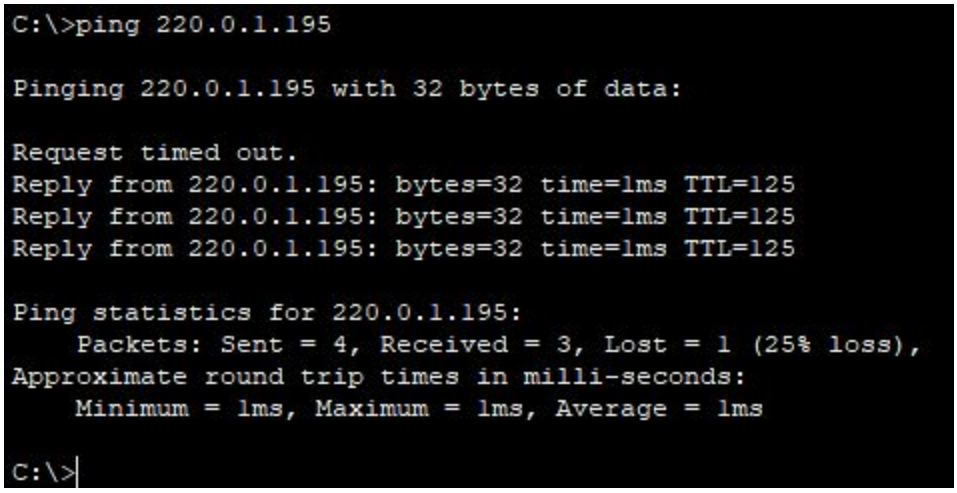
Ping statistics for 220.0.1.227:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 8ms

C:\>
```

SM_PC_2 (220.0.1.163) to HTTP Server (220.0.1.218)



SM_PC_2 (220.0.1.163) to A_PC_2 (220.0.1.195)



RS_PC_3 (220.0.1.228) to RD_PC_3 (220.0.1.68)

```
Packet Tracer PC Command Line 1.0
C:\>ping 220.0.1.68

Pinging 220.0.1.68 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.68: bytes=32 time=14ms TTL=126
Reply from 220.0.1.68: bytes=32 time=13ms TTL=126
Reply from 220.0.1.68: bytes=32 time=2ms TTL=126

Ping statistics for 220.0.1.68:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 14ms, Average = 9ms
```

RS_PC_3 (220.0.1.228) to SM_PC_3 (220.0.1.164)

```
C:\>ping 220.0.1.164

Pinging 220.0.1.164 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.164: bytes=32 time=10ms TTL=126
Reply from 220.0.1.164: bytes=32 time=2ms TTL=126
Reply from 220.0.1.164: bytes=32 time=2ms TTL=126

Ping statistics for 220.0.1.164:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 4ms

C:\>
```

RS_PC_3 (220.0.1.228) to HTTP Server (220.0.1.218)

```
C:\>ping 220.0.1.218

Pinging 220.0.1.218 with 32 bytes of data:

Reply from 220.0.1.218: bytes=32 time=1ms TTL=126
Reply from 220.0.1.218: bytes=32 time=2ms TTL=126
Reply from 220.0.1.218: bytes=32 time=11ms TTL=126
Reply from 220.0.1.218: bytes=32 time=1ms TTL=126

Ping statistics for 220.0.1.218:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 3ms

C:\>
```

RS_PC_3 (220.0.1.228) to A_PC_3 (220.0.1.196)

```
C:\>ping 220.0.1.196

Pinging 220.0.1.196 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.196: bytes=32 time=2ms TTL=124
Reply from 220.0.1.196: bytes=32 time=11ms TTL=124
Reply from 220.0.1.196: bytes=32 time=11ms TTL=124

Ping statistics for 220.0.1.196:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 11ms, Average = 8ms

C:\>
```

A_PC_4 (220.0.1.197) to RD_PC_4 (220.0.1.69)

```
C:\>ping 220.0.1.69

Pinging 220.0.1.69 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.69: bytes=32 time=2ms TTL=125
Reply from 220.0.1.69: bytes=32 time=2ms TTL=125
Reply from 220.0.1.69: bytes=32 time=18ms TTL=125

Ping statistics for 220.0.1.69:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 18ms, Average = 7ms

C:\>
```

A_PC_4 (220.0.1.197) to RS_PC_4 (220.0.1.229)

```
C:\>ping 220.0.1.229

Pinging 220.0.1.229 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.229: bytes=32 time=15ms TTL=124
Reply from 220.0.1.229: bytes=32 time=2ms TTL=124
Reply from 220.0.1.229: bytes=32 time=2ms TTL=124

Ping statistics for 220.0.1.229:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 15ms, Average = 6ms

C:\>
```

A_PC_4 (220.0.1.197) to HTTP Server (220.0.1.218)

```
C:\>ping 220.0.1.218

Pinging 220.0.1.218 with 32 bytes of data:

Reply from 220.0.1.218: bytes=32 time=3ms TTL=125
Reply from 220.0.1.218: bytes=32 time=2ms TTL=125
Reply from 220.0.1.218: bytes=32 time=1ms TTL=125
Reply from 220.0.1.218: bytes=32 time=2ms TTL=125

Ping statistics for 220.0.1.218:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>
```

A_PC_4 (220.0.1.197) to SM_PC_4 (220.0.1.165)

```
C:\>ping 220.0.1.165

Pinging 220.0.1.165 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.165: bytes=32 time=12ms TTL=125
Reply from 220.0.1.165: bytes=32 time=1ms TTL=125
Reply from 220.0.1.165: bytes=32 time=3ms TTL=125

Ping statistics for 220.0.1.165:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 5ms

C:\>
```

HTTP Server (220.0.1.218) to RD_PC_5 (220.0.1.70)

```
C:\>ping 220.0.1.70

Pinging 220.0.1.70 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.70: bytes=32 time=11ms TTL=127
Reply from 220.0.1.70: bytes=32 time=1ms TTL=127
Reply from 220.0.1.70: bytes=32 time=37ms TTL=127

Ping statistics for 220.0.1.70:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 37ms, Average = 16ms

C:\>
```

HTTP Server (220.0.1.218) to RS_PC_5 (220.0.1.230)

```
C:\>ping 220.0.1.230

Pinging 220.0.1.230 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.230: bytes=32 time=1ms TTL=126
Reply from 220.0.1.230: bytes=32 time=1ms TTL=126
Reply from 220.0.1.230: bytes=32 time=2ms TTL=126

Ping statistics for 220.0.1.230:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>
```

HTTP Server (220.0.1.218) to SM_PC_5 (220.0.1.166)

```
C:\>ping 220.0.1.166

Pinging 220.0.1.166 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.166: bytes=32 time<1ms TTL=127
Reply from 220.0.1.166: bytes=32 time<1ms TTL=127
Reply from 220.0.1.166: bytes=32 time<1ms TTL=127

Ping statistics for 220.0.1.166:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

HTTP Server (220.0.1.218) to A_PC_5 (220.0.1.198)

```
C:\>ping 220.0.1.198

Pinging 220.0.1.198 with 32 bytes of data:

Request timed out.
Reply from 220.0.1.198: bytes=32 time=13ms TTL=125
Reply from 220.0.1.198: bytes=32 time=1ms TTL=125
Reply from 220.0.1.198: bytes=32 time=3ms TTL=125

Ping statistics for 220.0.1.198:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 13ms, Average = 5ms

C:\>
```

Note: We have included the ping testing results in the form of screenshots as Cisco Packet Tracer does not support copying the output of ping results to a TFTP Server

Appendix B: Routing Table

AR1 Routing Table

```
AR1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     2.0.0.0/32 is subnetted, 1 subnets
C       2.1.1.1/32 is directly connected, Loopback0
     200.200.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.200.100.0/30 is directly connected, Serial0/0/0
L       200.200.100.1/32 is directly connected, Serial0/0/0
     220.0.1.0/24 is variably subnetted, 10 subnets, 6 masks
O       220.0.1.64/26 [110/65] via 220.0.1.250, 00:00:04, Serial0/0/1
O       220.0.1.160/27 [110/65] via 220.0.1.250, 00:00:04, Serial0/0/1
O       220.0.1.208/29 [110/65] via 220.0.1.250, 00:00:04, Serial0/0/1
O       220.0.1.216/29 [110/65] via 220.0.1.250, 00:00:04, Serial0/0/1
O       220.0.1.224/28 [110/129] via 220.0.1.250, 00:00:04, Serial0/0/1
C       220.0.1.240/30 is directly connected, GigabitEthernet0/0
L       220.0.1.241/32 is directly connected, GigabitEthernet0/0
C       220.0.1.248/30 is directly connected, Serial0/0/1
L       220.0.1.249/32 is directly connected, Serial0/0/1
O       220.0.1.252/30 [110/128] via 220.0.1.250, 00:00:04, Serial0/0/1
S*   0.0.0.0/0 is directly connected, Serial0/0/0

AR1#
```

AR2 Routing Table

```
AR2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 220.0.1.241 to network 0.0.0.0

     2.0.0.0/32 is subnetted, 1 subnets
C       2.2.2.2/32 is directly connected, Loopback0
     220.0.1.0/24 is variably subnetted, 11 subnets, 6 masks
O       220.0.1.64/26 [110/66] via 220.0.1.241, 00:00:09, GigabitEthernet0/0
O       220.0.1.160/27 [110/66] via 220.0.1.241, 00:00:09, GigabitEthernet0/0
C       220.0.1.192/28 is directly connected, GigabitEthernet0/1
L       220.0.1.193/32 is directly connected, GigabitEthernet0/1
O       220.0.1.208/29 [110/66] via 220.0.1.241, 00:00:09, GigabitEthernet0/0
O       220.0.1.216/29 [110/66] via 220.0.1.241, 00:00:09, GigabitEthernet0/0
O       220.0.1.224/28 [110/130] via 220.0.1.241, 00:00:09, GigabitEthernet0/0
C       220.0.1.240/30 is directly connected, GigabitEthernet0/0
L       220.0.1.242/32 is directly connected, GigabitEthernet0/0
O       220.0.1.248/30 [110/65] via 220.0.1.241, 00:00:09, GigabitEthernet0/0
O       220.0.1.252/30 [110/129] via 220.0.1.241, 00:00:09, GigabitEthernet0/0
O*E2 0.0.0.0/0 [110/1] via 220.0.1.241, 00:00:09, GigabitEthernet0/0
```

RSB1 Routing Table

```
RSB1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 220.0.1.254 to network 0.0.0.0

     220.0.1.0/24 is variably subnetted, 11 subnets, 6 masks
O       220.0.1.64/26 [110/65] via 220.0.1.254, 00:01:22, Serial0/0/0
O       220.0.1.160/27 [110/65] via 220.0.1.254, 00:01:22, Serial0/0/0
O       220.0.1.192/28 [110/130] via 220.0.1.254, 00:00:42, Serial0/0/0
O       220.0.1.208/29 [110/65] via 220.0.1.254, 00:01:22, Serial0/0/0
O       220.0.1.216/29 [110/65] via 220.0.1.254, 00:01:22, Serial0/0/0
C       220.0.1.224/28 is directly connected, GigabitEthernet0/0
L       220.0.1.225/32 is directly connected, GigabitEthernet0/0
O       220.0.1.240/30 [110/129] via 220.0.1.254, 00:00:42, Serial0/0/0
O       220.0.1.248/30 [110/128] via 220.0.1.254, 00:01:22, Serial0/0/0
C       220.0.1.252/30 is directly connected, Serial0/0/0
L       220.0.1.253/32 is directly connected, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 220.0.1.254, 00:01:12, Serial0/0/0

RSB1#
```

MBR1 Routing Table

```
MBR1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 220.0.1.249 to network 0.0.0.0

     3.0.0.0/32 is subnetted, 1 subnets
C       3.3.3.3/32 is directly connected, Loopback0
     220.0.1.0/24 is variably subnetted, 15 subnets, 6 masks
C       220.0.1.64/26 is directly connected, GigabitEthernet0/0.10
L       220.0.1.65/32 is directly connected, GigabitEthernet0/0.10
C       220.0.1.160/27 is directly connected, GigabitEthernet0/0.20
L       220.0.1.161/32 is directly connected, GigabitEthernet0/0.20
O       220.0.1.192/28 [110/66] via 220.0.1.249, 00:01:22, Serial0/0/1
C       220.0.1.208/29 is directly connected, GigabitEthernet0/0.99
L       220.0.1.209/32 is directly connected, GigabitEthernet0/0.99
C       220.0.1.216/29 is directly connected, GigabitEthernet0/0.30
L       220.0.1.217/32 is directly connected, GigabitEthernet0/0.30
O       220.0.1.224/28 [110/65] via 220.0.1.253, 00:02:02, Serial0/0/0
O       220.0.1.240/30 [110/65] via 220.0.1.249, 00:01:22, Serial0/0/1
C       220.0.1.248/30 is directly connected, Serial0/0/1
L       220.0.1.250/32 is directly connected, Serial0/0/1
C       220.0.1.252/30 is directly connected, Serial0/0/0
L       220.0.1.254/32 is directly connected, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 220.0.1.249, 00:02:02, Serial0/0/1

MBR1#
```

Note: We have included the routing tables from each routers in the form of screenshots as Cisco Packet Tracer does not support copying the output of the routing table to a TFTP Server

Appendix C: Pricing References

Links for pricing of physical layer equipment

Most equipment cost are in USD so we converted it to SGD (Shipping cost not included)

- For 2960-24TT switches:
  - https://www.cnet.com/products/cisco-catalyst-2960-24tt-switch-24-ports-managed-rack-mountable-series/
  - https://www.router-switch.com/Price-cisco-switches-cisco-switch-catalyst-2960_c19
  - https://www.amazon.com/Cisco-WS-C2960-24TT-L-Catalyst-2960-24TT-Ethernet/dp/B008D719KM/ref=sr_1_11?dchild=1&keywords=Cisco+WS-C2960-24TT-L&qid=1603954759&refinements=p_89%3ACisco&rnid=2528832011&s=electronics&sr=1-11

- For 2901 routers:
  - https://www.amazon.sg/Cisco-CISCO2901-Integrated-Services-Router/dp/B0030E9EVE
  - https://www.router-switch.com/cisco2901-k9-p-155.html
  - https://www.secureitstore.com/2901.asp
  - https://www.router-switch.com/search/HWIC-2T (Serial Interface)

- For ethernet cable:
  - https://www.primuscable.com/collections/cat8-bulk-ethernet-cable/products/cat8-v2-bulk-ethernet-cable-40g#whats-in-the-box (Solid Core)
  - https://www.amazon.com/WeCable-Shielded-Stranded-40Gbps-Ethernet/dp/B07HB5TFRT (stranded)

- For 6/24 ports patch panel CAT8:
  - https://www.amazon.com/Yankok-Unshielded-Compatible-Networks-YKPP006/dp/B07WHZB6P9/ref=sr_1_5?dchild=1&keywords=cat6+patch+panel+6+port&sr=8-5
  - https://warehousecables.com/store/products/patch-panel-24-port-cat8-shielded-110-type-568a-b

- RJ45 jack socket/connector:
  - https://sg.rs-online.com/web/p/ethernet-connectors/3316386/
  - https://www.amazon.com/Ethernet-RJ45-Connector-Modular-Connectors/dp/B0771MCFQ7

- Wall port
  - https://www.amazon.com/Certicable-Stainless-Single-Network-Shielded/dp/B07CBNR92K

    - https://www.newegg.com/certicable-cj-9a09-xx68/p/1GY-0013-000K4

- Cable tie
    - https://www.amazon.sg/Tensile-Strength-Durable-Outdoor-Resistant/dp/B07545HGR5/ref=asc_df_B07545HGR5/?tag=googleshoppin-22&linkCode=df0&hvadid=404125714497&hvpos=&hvnetw=g&hvrand=14232521983355498177&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9062548&hvtargid=pla-358250759038&psc=1

- PVC pipe
    - https://uk.rs-online.com/web/p/pvc-pipes-plastic-pipes/2123953/

- Rack
    - https://www.cableorganizer.com/computer-cabinets/4-post-open-frame-adjustable-rack.html

Generated Physical Diagrams:

- https://drive.google.com/file/d/1_6wt29F2kM0BfR5liemQv5cjpk2_BPv9/view?usp=sharing

This is a google drive link to the draw.io file containing our physical layer diagrams. This is only accessible by email accounts within the “Ngee Ann Polytechnic” Group.